

Program1

```
manchodu@Srini:~$ cd BSLAN
-bash: cd: BSLAN: No such file or directory
manchodu@Srini:~$ cd ~/BSLAB
manchodu@Srini:~/BSLAB$ ls
fork  pork  program  program.c  program1  program1.c
manchodu@Srini:~/BSLAB$ cat program1.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
manchodu@Srini:~/BSLAB$ ./program1
Hello
Hello
manchodu@Srini:~/BSLAB$ |
```

Program2

```
@00-@0x  ◆3◆◆
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
manchodu@Srini:~/BSLAB$ ./program2
Hello
Hello
manchodu@Srini:~/BSLAB$ |
```

Program3

```
manchodu@Srini:~/BSLAB$ nano program3.c
manchodu@Srini:~/BSLAB$ gcc program3.c -o program3
manchodu@Srini:~/BSLAB$ ./program3
I am Parent
I am Child
manchodu@Srini:~/BSLAB$ cat program3.c
#include <stdio.h>
#include <unistd.h>
int main()
{
pid_t pid;
pid = fork();
if(pid == 0)
printf("I am Child\n");
else
printf("I am Parent\n");
return 0;
}
manchodu@Srini:~/BSLAB$ |
```

Program4

```
manchodu@Srini:~/BSLAB$ nano program4.c
manchodu@Srini:~/BSLAB$ gcc program4.c -o program4
manchodu@Srini:~/BSLAB$ ./program4
Parent PID = 523
Child PID = 524
Child PID = 524
Parent PID = 523
manchodu@Srini:~/BSLAB$ cat program4.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid;
    pid = fork();
    if(pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }
    return 0;
}
manchodu@Srini:~/BSLAB$ |
```

Program5

```
manchodu@Srini:~/BSLAB$ nano program5.1.c
manchodu@Srini:~/BSLAB$ gcc program5.1.c -o program5.1
manchodu@Srini:~/BSLAB$ ./program5
Child Executing
Parent Executing
manchodu@Srini:~/BSLAB$ cat program5.c
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child Executing\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Executing\n");
    }
    return 0;
}
manchodu@Srini:~/BSLAB$ |
```

Program6

```
manchodu@Srini:~/BSLAB$ nano program6.1.c
manchodu@Srini:~/BSLAB$ gcc program6.1.c -o program6.1
manchodu@Srini:~/BSLAB$ ./program6.1
PID = 572
PID = 573
PID = 575
PID = 574
manchodu@Srini:~/BSLAB$ cat program6.c
#include <stdio.h>
#include <unistd.h>
int main()
{
fork();
fork();
printf("PID = %d\n", getpid());
return 0;
}
How many processes are created?
manchodu@Srini:~/BSLAB$ |
```

Program 7

```

manchodu@Srini:~/BSLAB$ nano program7.c
manchodu@Srini:~/BSLAB$ gcc program7.c -o program7
manchodu@Srini:~/BSLAB$ ./program7
Parent Waiting...
Child executing ls...
manchodu@Srini:~/BSLAB$ total 192
drwxr-xr-x 2 manchodu manchodu 4096 Jul 29 03:28 fork
drwxr-xr-x 2 manchodu manchodu 4096 Jul 29 03:28 pork
-rwxr-xr-x 1 manchodu manchodu 16008 Jul 29 03:32 program
-rw-r--r-- 1 manchodu manchodu 98 Jul 29 03:31 program.c
-rwxr-xr-x 1 manchodu manchodu 16008 Jul 29 03:33 program1
-rw-r--r-- 1 manchodu manchodu 94 Jul 29 03:33 program1.c
-rwxr-xr-x 1 manchodu manchodu 16008 Jul 30 09:54 program2
-rw-r--r-- 1 manchodu manchodu 98 Jul 30 09:56 program2.c
-rwxr-xr-x 1 manchodu manchodu 16008 Jul 30 09:57 program3
-rw-r--r-- 1 manchodu manchodu 168 Jul 30 09:56 program3.c
-rwxr-xr-x 1 manchodu manchodu 16096 Jul 30 09:58 program4
-rw-r--r-- 1 manchodu manchodu 282 Jul 30 09:58 program4.c
-rwxr-xr-x 1 manchodu manchodu 16048 Jul 30 09:59 program5
-rwxr-xr-x 1 manchodu manchodu 16048 Jul 30 10:00 program5.1
-rw-r--r-- 1 manchodu manchodu 220 Jul 30 10:00 program5.1.c
-rw-r--r-- 1 manchodu manchodu 199 Jul 30 09:59 program5.c
-rwxr-xr-x 1 manchodu manchodu 16048 Jul 30 10:02 program6.1
-rw-r--r-- 1 manchodu manchodu 121 Jul 30 10:01 program6.1.c
-rw-r--r-- 1 manchodu manchodu 153 Jul 30 10:01 program6.c
-rwxr-xr-x 1 manchodu manchodu 16048 Jul 30 10:03 program7
-rw-r--r-- 1 manchodu manchodu 225 Jul 30 10:03 program7.c

manchodu@Srini:~/BSLAB$ cat program7.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child executing ls...\n");
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
    {
        printf("Parent Waiting...\n");
    }
    return 0;
}
manchodu@Srini:~/BSLAB$ |

```

```
manchodu@Srini:~/BSLAB$ nano program8.c
manchodu@Srini:~/BSLAB$ gcc program8.c -o program8
manchodu@Srini:~/BSLAB$ ./program8
PID = 608
PID = 611
PID = 609
PID = 612
PID = 613
PID = 607
PID = 610
PID = 614
manchodu@Srini:~/BSLAB$ cat program8.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    int i;
    for(i=0;i<3;i++)
        fork();
    printf("PID = %d\n",getpid());
    return 0;
}
manchodu@Srini:~/BSLAB$ |
```

Program 10